

Huihuo Zheng

Computer Scientist

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CURRICULUM VITAE

(June 1st, 2026)

Education

- PhD (2016), Physics, University of Illinois at Urbana-Champaign, Urbana, IL
- MPhil (2010), Physics, Hong Kong University of Science and Technology, Hong Kong
- BS (2008), Physics, Tsinghua University, Beijing, China

Professional Experience

- **Argonne National Laboratory**, Leadership Computing Facility Division (United States)
Computer Scientist (July 2023 – Present)
Assistant Computer Scientist (May 2019 – June 2023)
Postdoctoral Appointee (August 2010 – May 2019)
 - Co-leading American Science Cloud Model Services, delivering experiment tracking, model services, database / inference / workflow as a service system in AmSC
 - Led research on Agentic AI for autonomous HPC
 - Led the development of AskALCF – an agentic chatbot system for HPC user support
 - Developed and trained advanced deep learning models, including large language models, tailored for scientific research applications.
 - Led innovative research initiatives focused on optimizing data management for artificial intelligence workflows.
 - Delivered comprehensive lectures and training sessions on artificial intelligence, deep learning, and high-performance computing for diverse groups, including scientists, researchers, and students.
 - Actively supported and mentored academic users conducting cutting-edge research using Argonne's high-performance computing resources.
 - Served as Co-chair for the MLPerf Storage Working Group, overseeing benchmark development, establishing benchmark submission guidelines, and driving community standards for evaluating storage performance in AI applications.
- **Wheaton College**, Department of Math and Computer Science (United States)
Adjunct Instructor (Aug 2025 - Present)
 - CSCI234 Python Programming in Business Analytics
 - CSCI394 High Performance Computing and AI

Honors and Awards

- Golden Bell Finalist, 2025
- Impact Argonne Award for ALCF AI for Golden Bell Finalist, 2024

- Golden Bell Finalist, 2024
- Impact Argonne Award for ALCF AI for Science Training, 2023
- Impact Argonne Award for ALCF AI for Science Training, 2022
- Best Paper Award, IEEE/ACM 21st International Symposium on Cluster, Cloud and Internet Computing, 2021
- Editors' Suggestions, Physics Review Letters, 2017
- Outstanding Poster Presentation Award, Argonne National Laboratory, 2017
- University Fellowship for outstanding research, University of Illinois at Urbana-Champaign, 2015
- Scott Anderson Award for outstanding teaching assistant, University of Illinois at Urbana-Champaign, 2012
- Chi-Sun YEH Prize, for academic excellence in physics, Tsinghua University, 2008
- Excellent Graduate Honor, Tsinghua University, 2008
- Excellent B.Sc. Thesis Award, Tsinghua University, 2008

Professional Activities

- Program Committee member for the 6th Workshop on Performance Engineering, Modelling, Analysis, and VisualizatiOn STrategy (PERMAVOST 2026)
- Program Committee member for 32nd International European Conference on Parallel and Distributed Computing 24-28 August 2026, Pisa, Italy
- SC25 Program Committee for ML Track, 2025
- Co-Chair for MLPerf Storage working group, leading the effort of development storage benchmarks for AI and defining benchmark submission rules and policies, 2022 – present.
- Mentor for INCITE GPU Hackathon, April 2022
- Panellist for 2022 OCP Global Summit: *Why Are We Talking About AI HW/SW Co-Design Again?*
- SC22 Planning Committee for research poster session, 2022
- Mentor for Nvidia GPU Hackathon, July 2022
- Symposium Organizer and Chair: Microscopy & Microanalysis Symposium 2020, Section A01 – Advances in Modeling, Simulation, and Machine Learning for Microscopy Data sets, August 2020
- Organizer and lecturer for ALCF Simulation, Data and Learning Workshop, October 2021
- Organizer and lecturer for ALCF Simulation, Data and Learning Workshop, October 2019
- Lecturer and reviewing committee for ATPESC, Argonne Training Program on Extreme-scale Computing, 2019-present
- Reviewer for DOE INCITE Computational Readiness Review
- Reviewer for DOE Advanced Scientific Computing Research (ASCR) Program
- Reviewer for DOE ASCR Leadership Computing Challenge (ALCC) Program
- Reviewer for journals: Physical Reviews Letters, Physical Reviews Research, Physical Reviews Materials, Physical Reviews Applied, Physical Reviews B, Physical Reviews E, Scientific Reports, Journal of Physics: Condensed Matter, Journal of Chemical Theory and Computation, 2D Materials, Microelectronic Journal, Open Physics, Molecular Simulation, Journal of Applied Physics

Professional Development

- Co-Chair of MLPerf storage working group, 2022 – present
- Member of OCP AI co-design working group, 2022 - present
- Member of IEEE, 2018 - present
- Member of ACM, 2018 – present
- Member of the American Physical Society, 2016 - 2020

Publications, Presentations, and Other Evidence of Achievement

Google Scholar profile: <https://scholar.google.com/citations?user=dd7fUtEAAAAJ&hl=en>

Refereed Journal Articles

- [1] Akhilesh Bondapalli, Huihuo Zheng, Oluwaseun T Ajayi, Murat Keceli, Haritha Siddabathuni Som, Taylor Childers, Lisa Childers, Yasaman Ghadar, Michael Papka, Venkatram Vishwanath, Rong Ge, *AskHPC: A ChatBot for High Performance Computing User Support*, Proceedings of the SC'25 Workshops of the International Conference for High Performance Computing, Networking, Storage and Analysis
- [2] Väinö Hatanpää, Eugene Ku, Jason Stock, Murali Emani, Sam Foreman, Chunyong Jung, Sandeep Madireddy, Tung Nguyen, Varuni Sastry, Ray AO Sinurat, Sam Wheeler, Huihuo Zheng, Troy Arcomano, Venkatram Vishwanath, Rao Kotamarthi, AERIS: Argonne Earth Systems Model for Reliable and Skillful Predictions, SC25, International Conference for High Performance Computing, Networking, Storage and Analysis
- [3] Benjamin S Allen, James Anchell, et al, Aurora: Architecting Argonne's First Exascale Supercomputer for Accelerated Scientific Discovery, arXiv:2509.08207
- [4] Wenhua Zuo, Wenhua Zuo, Huihuo Zheng, Tanjin He, Rick L. Stevens, Khalil Amine, Gui-Liang Xu, Large language models for batteries, Joule, Volume 9, Issue 8, 102037
- [5] Ozan Gokdemir, Carlo Siebensschuh, Alexander Brace, Azton Wells, Brian Hsu, Kyle Hippe, Priyanka Setty, Aswathy Ajith, J Gregory Pauloski, Varuni Sastry, Sam Foreman, Huihuo Zheng, Heng Ma, Bharat Kale, Nicholas Chia, Thomas Gibbs, Michael Papka, Thomas Brettin, Francis Alexander, Anima Anandkumar, Ian Foster, Rick Stevens, Venkatram Vishwanath, Arvind Ramanathan, HiPerRAG: High-Performance Retrieval Augmented Generation for Scientific Insights, 2025, Proceedings of the Platform for Advanced Scientific Computing Conference Pages 1-13
- [6] Scot Breitenfeld, Houjun Tang, Huihuo Zheng, Jordan Henderson, and Suren Byna. "HDF5 in the exascale era: Delivering efficient and scalable parallel I/O for exascale applications." *International Journal of High-Performance Computing Applications*, 39(1): 65–78 (2025).
- [7] Minyang Tian, E. A. Huerta, Huihuo Zheng, Prayush Kumar. "Physics-inspired spatiotemporal-graph AI ensemble for the detection of higher order wave mode signals of spinning binary black hole mergers." *Machine Learning: Science and Technology* 5, no. 2: 025056 (2024)
- [8] Jonathan Schwartz, Chris Harris, Jacob Pietryga, Huihuo Zheng, Prashant Kumar, Anastasiia Visheratina, Nicholas A. Kotov, Brianna Major, Patrick Avery, Peter Ercius, Utkarsh Ayachit, Berk Geveci, David A. Muller, Alessandro Genova, Yi Jiang, Marcus Hanwell, and Robert Hovden. "Imaging 3D Chemistry at 1 nm Resolution with Fused Multi-Modal Electron Tomography." *Nature Communications* 15, no. 1: 47558 (2024)
- [9] Jean Luca Bez, Houjun Tang, M. Scot Breitenfeld, Huihuo Zheng, Wei-Keng Liao, Kaiyuan Hou, Zanhua Huang, and Suren Byna. "h5bench: A unified benchmark suite for evaluating HDF5 I/O performance on pre-exascale platforms." *Concurrency and Computation: Practice and Experience*, 36(16): e8046 (2024).
- [10] Jonathan Schwartz, Chris Harris, Jacob Pietryga, Huihuo Zheng, Prashant Kumar, Anastasiia Visheratina, Nicholas A. Kotov, et al. "Real-Time 3D Analysis during Electron Tomography Using Tomviz." *Nature Communications* 13, no. 1, 4458 (2022)
- [11] Pranshu Chaturvedi, Asad Khan, Minyang Tian, Eliu. A. Huerta, and Huihuo Zheng, "Inference-optimized AI and high-performance computing for gravitational wave detection at scale", *Frontiers in Artificial Intelligence*. 5:828672 (2022)
- [12] Asad Khan, Eliu A. Huerta, and Huihuo Zheng, "Interpretable AI forecasting for numerical relativity waveforms of quasicircular, spinning, nonprecessing binary black hole mergers", *Phys. Rev. D* 105, 024024 (2022)
- [13] Jin-Ke Bao, Huihuo Zheng, Jianguo Wen, Sitaram Ramakrishnan, Hao Zheng, Jidong Samuel Jiang, Daniel Bugaris, Gang Cao, Duck Young Chung, Sander van Smaalen, Mercouri G

- Kanatidis, Superconductivity in Y4RuGe8 with a Vacancy-Ordered CeNiSi2-Type Superstructure, *Chem. Mater.* 33, 19, 7839–7847 (2021)
- [14] Jin-Ke Bao, Daniel Bugaris, Huihuo Zheng, Duck Young Chung and Mercouri Kanatzidis. “A no centrosymmetric polymorph of LuRuGe.” *Inorganic Chemistry* 60, no. 11, 7827- 7833 (2021)
 - [15] Jonathan Schwartz, Huihuo Zheng, Marcus Hanwell, Yi Jiang and Robert Hovden. “Dynamic Compressed Sensing for Real-Time Tomographic Reconstruction.” *Ultramicroscopy* 219 (2020)
 - [16] Huihuo Zheng, Marco Govoni, and Giulia Galli, “Dielectric dependent hybrid functional for interfaces”, *Phys. Rev. Materials* 3, 073803 (2019).
 - [17] Asad Khan, Eliu A. Huerta, Sibo Wang, Robert Gruendl, Elise Jennings, Huihuo Zheng, “Unsupervised Learning and Data Clustering for the Construction of Galaxy Catalogs at Scale in the Dark Energy Survey”, *Physics Letters B* 795, 248-258 (2019).
 - [18] Jin-Ke Bao, Daniel E. Bugaris, Huihuo Zheng, Kristin Willa, Ulrich Welp, Duck Young Chung, Mercouri G. Kanatzidis, “Superconductivity in Y7Ru4InGe12”, *Phys. Rev. Materials* 3, 024802 (2019).
 - [19] Huihuo Zheng, Hitesh J. Changlani, Kiel T. Williams, Brian Busemeyer, and Lucas K. Wagner, “From real materials to model Hamiltonians with density matrix downfolding”, *Frontiers in Physics*, 6, 43 (2018) **(Invited)**
 - [20] Huihuo Zheng, Yu Gan, Peter Abbamonte, and Lucas K. Wagner, “Importance of σ bonding electrons for the accurate description of electron correlation in graphene”, *Phys. Rev. Lett.* 119 (16), 166402 (2017) **(Editors’ Suggestion)**
 - [21] Huihuo Zheng, and Lucas K. Wagner, “Computation of the correlated metal-insulator transition in vanadium dioxide from first principles”, *Phys. Rev. Lett.* 114 (17), 176401 (2015).
 - [22] Hitesh J. Changlani, Huihuo Zheng, and Lucas K. Wagner, “Density-matrix based determination of low- energy model Hamiltonians from ab initio wavefunctions”, *J. Chem. Phys.* 143, 102814 (2015).
 - [23] Xueqing Huang, Yun Lai, Zhi-Hong. Hang, Huihuo Zheng, and Che Ting Chan, “Dirac cones induced by accidental degeneracy in photonic crystals and zero-refractive-index materials”, *Nature Materials* 10 (8), 582-586 (2011).
 - [24] Shubo Wang, Huihuo Zheng, Jun-Jun Xiao, Zhifang Lin, and Che Ting Chan. “Fast multipole boundary element method for three-dimensional electromagnetic scattering problem”, *Int. J. Comput. Mater. Sci. Eng.* 01, 04 (2012).
 - [25] Shubo Wang, Jack Ng, Hui Liu, Huihuo Zheng, Zhihong Hang, Che Ting Chan, “Sizable electromagnetic forces in parallel-plate metallic cavity”, *Phys. Rev. B* 84(7), 075114 (2012).
 - [26] Yu-Ping Kuang, Jian-Ping Ma, Otto Nachtmann, Wan-Peng Xie, and Hui-Huo Zheng. “Testing the anomalous color-electric dipole moment of the c quark from $\psi' \rightarrow J/\psi + \pi^+ + \pi^-$ at the Beijing Spectrometer”, *Phys. Rev. D* 85(11), 114010 (2012).
 - [27] Jian-Wen Dong, Huihuo Zheng, Yun Lai, He-Zhou Wang, and Che Ting Chan, “Metamaterial slab as a lens, a cloak, or an intermediate”, *Phys. Rev. B* 83, 115124 (2011).
 - [28] Yun Lai, Huihuo Zheng, Zhao-Qing Zhang and Che Ting Chan, “Manipulating sources using transformation optics with folded geometry”, *J. Opt.* 13, 024009 (2011).
 - [29] Huihuo Zheng, Jun-Jun Xiao, Yun Lai, and Che Ting Chan “Exterior optical cloaking and illusions by using active sources: a boundary element perspective”, *Phys. Rev. B* 81, 195116 (2010).
 - [30] Jun-Jun Xiao, Huihuo Zheng, Yunxu Sun, and Yong Yao, “Bipolar optical forces on dielectric and metallic nanoparticles by evanescent wave”, *Opt. Lett.* 35, 962-964 (2010).

Refereed publication in conference or workshop proceedings/reports

- [1] Noah Lewis, Kaushik Velusamy, Kevin Harms, and Huihuo Zheng. “Copper: Cooperative Caching Layer for Scalable Data Loading in Exascale Supercomputers.” *International Conference for High Performance Computing, Networking, Storage and Analysis (SC) Workshops*, pp. 1320–1329 (2024).

- [2] Gautham Dharuman, Kyle Hippe, Alexander Brace, Sam Foreman, Väinö Hatanpää, Varuni K Sastry, Huihuo Zheng, Logan Ward, Servesh Muralidharan, Archit Vasan, Bharat Kale, Carla M Mann, Heng Ma, Yun-Hsuan Cheng, Yuliana Zamora, Shengchao Liu, Chaowei Xiao, Murali Emani, Tom Gibbs, Mahidhar Tatineni, Deepak Canchi, Jerome Mitchell, Koichi Yamada, Maria Garzaran, Michael E Papka, Ian Foster, Rick Stevens, Anima Anandkumar, Venkatram Vishwanath, Arvind Ramanathan, MProt-DPO: Breaking the ExaFLOPS Barrier for Multimodal Protein Design Workflows with Direct Preference Optimization, International Conference for High Performance Computing, Networking, Storage and Analysis, 2024
- [3] Hariharan Devarajan, Loïc Pottier, Kaushik Velusamy, Huihuo Zheng, Izzet Yildirim, Olga Kogiou, Weikuan Yu, Anthony Kougkas, Xian-He Sun, Jae-Seung Yeom, and Kathryn M. Mohror. "DFTracer: An Analysis-Friendly Data Flow Tracer for AI-Driven Workflows." *International Conference for High Performance Computing, Networking, Storage and Analysis (SC)*, p. 17 (2024).
- [4] Jason Manassa, Jonathan Schwartz, Jaewhan Oh, Zichao Wendy Di, Yi Jiang, Huihuo Zheng, Jeffrey A Fessler, Yongsoo Yang, Robert Hovden, Recovering Stoichiometry via Multi-Modal Fused Electron Tomography, Microscopy and Microanalysis, Volume 30, Supplement S1, July 2024, 126
- [5] Robert Hovden, Jonathan Schwartz, Suk Hyun Sung, Zichao Wendy Di, Yi Jiang, Jason Manassa, Jacob Pietryga, Yiwen Qian, Min Gee Cho, Jonathan L Rowell, Huihuo Zheng, Richard D Robinson, Junsu Gu, Alexey Kirilin, Steve Rozeveld, Peter Ercius, Mary Scott, "Chemical Electron Tomography at Lower Dose and Higher Resolution", Volume 30, Supplement S1, July 2024, 890
- [6] Huihuo Zheng, Venkatram Vishwanath, Quincey Koziol, Houjun Tang, and Suren Byna, "Cache VOL: Efficient Parallel I/O through Caching Data on the Node-local Storage", The 22nd IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing, 2022
- [7] Hariharan Devarajan, Anthony Kougkas, Huihuo Zheng, Xian-He Sun and Venkatram Vishwanath, "Stimulus: Accelerate Data Management for Scientific AI applications in HPC", The 22nd IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing, 2022 (**Best Paper Award**)
- [8] Jonathan Schwartz, Chris Harris, Jacob Pietryga, Huihuo Zheng, Prashant Kumar, Anastasiia Visheratina, Nicholas Kotov, Yi Jiang, Marcus Hanwell, Robert Hovden, "Real-Time 3D Analysis During Tomography Experiments on tomviz", Microscopy and Microanalysis, Volume 27, Supplement S1, August 2021, pp. 2860 – 2862
- [9] Jonathan Schwartz, Yi Jiang, Zichao Wendy Di, Tao Ma, Huihuo Zheng, Steve Rozeveld, Robert Hovden, "Recovering Chemistry at Atomic Resolution using Multi-Modal Spectroscopy", Microscopy and Microanalysis, Volume 27, Supplement S1, August 2021, pp. 1226 – 1228
- [10] Hariharan Devarajan, Huihuo Zheng, Anthony Kougkas, Xian-He Sun and Venkatram Vishwanath, "DLIO: A Data- Centric Benchmark for Scientific Deep Learning Applications," 2021 IEEE/ACM 21st International Symposium on Cluster, Cloud and Internet Computing (CCGrid), 2021, pp. 81-91.
- [11] Hariharan Devarajan, Huihuo Zheng, Xian-He Sun and Venkatram Vishwanath. "Understanding I/O behavior of Scientific Deep Learning Applications in HPC systems." 2020 International Conference for High Performance Computing, Networking, Storage, and Analysis, November 15-20, 2020.
- [12] Jonathan Schwartz, Yi Jiang, Zichao Wendy Di, Tao Ma, Huihuo Zheng, Steve Rozeveld, Robert Hovden, "Recovering Chemistry at Atomic Resolution using Multi-Modal Spectroscopy", Microscopy and Microanalysis, Volume 27, Supplement S1, August 2021, pp. 1226 – 1228
- [13] Reed Yalisove, Suk Hyun Sung, Jonathan Schwartz, Catherine Groschner, Philipp Pelz, Huihuo Zheng, Yi Jiang, Colin Ophus, Mary Scott, Peter Ercius, and Robert Hovden, "Achieving High-

- resolution of Large Specimens Using Aberration-corrected Tomography”, Microscopy and Microanalysis, Volume 26, Supplement S2, August 2020, pp. 1860 – 1862
- [14] Colleen Bertoni, JaeHyuk Kwack, Thomas Applencourt, Yasaman Ghadar, Brian Homerding, Christopher Knight, Brice Videau, Huihuo Zheng, Vitali Morozov, Scott Parker, Performance Portability Evaluation of OpenCL Benchmarks across Intel and NVIDIA Platforms, 2020 IEEE International Parallel and Distributed Processing Symposium Workshops (IPDPSW), 2020, pp. 330-339.
 - [15] Janet Knowles, Joseph A. Insleya, Silvio Rizzi, Elise Jennings, Asad Khan, E. A. Huerta, Sibo Wang, Robert Gruendl, Huihuo Zheng, “Visualizing Deep Learning at Scale for the Construction of Galaxy Catalogs in the Dark Energy Survey”, SC19 conference.
 - [16] JaeHyuk Kwack, Thomas Applencourt, Colleen Bertoni, Yasaman Ghadar, Huihuo Zheng, Christopher Knight, and Scott Parker, “Roofline-based Performance Efficiency of HPC Benchmarks and Applications on Current Generation of Processor Architectures”, Cray User Group Workshop 2019.
 - [17] Che Ting Chan, Xueqin Huang, Yun Lai, Zhi Hong Hang, and Huihuo Zheng, “Dirac cone and double zero materials”, AIP Conference Proceedings 1398, 21, 2011

Technical Reports

- [1] Huihuo Zheng, Christopher Knight, Giulia Galli, Marco Govoni, and Francois Gygi, “First-Principles Simulations of Functional Materials for Energy Conversion”, ALCF-3 Theta Early Science technical report, 2017. 2

Other publications/documents

- [1] HDF5 Cache VOL Documentation: <https://vol-cache.readthedocs.io/en/latest/>
- [2] Deep Learning I/O Benchmark (DLIO) Documentation: https://argonne-lcf.github.io/dlio_benchmark/
- [3] MLPerf Storage Working group Documentation: <https://github.com/mlcommons/storage/blob/main/README.md>

Invited talks presented at major conferences and symposia

- [1] “I/O Profiling and Benchmarking for AI Applications,” SC24, November 2024
- [2] Panel discussion: *Why Are We Talking About AI HW/SW Co-Design Again?* 2022 OCP Global Summit.

Seminars and colloquia presented

- [1] “First-principles Quantum Monte Carlo study of correlated electronic systems”, Oak Ridge National Laboratory, Knoxville, TN, March 2016
- [2] “Topological unidirectional waveguide”, Soochow University, Soochow, China, August 2012
- [3] “Metamaterial slab as a lens, a cloak, or an intermediate”, Soochow University, Soochow, China, May 2010

Contributed conference and other presentations

- [1] “MLPerf Storage Benchmark”, Intel storage working group meeting, March 2023
- [2] “HDF5: Pre-ECP, Present, and Future”, 2023 DOE Exascale Computing Project Annual Meeting, April 2023
- [3] “Distributed Deep Learning”, ALCF Simulation, Data and Learning Workshop, October 2022
- [4] “Deep Learning I/O Benchmarks”, MLPerf Storage Working Group Meeting, June 3, 2022
- [5] “HDF5 Cache VOL: Efficient parallel I/O through caching data on node-local storage”, CCGrid2022, The 22nd IEEE/ACM International Symposium on Cluster, Cloud and Internet Computing, May 2022

- [6] "Sharing Experience on HDF5 VOL Connectors Development and Maintenance", 2022 DOE Exascale Computing Project Annual Meeting, April 2022
- [7] "Using Parallel HDF5 Efficiently on HPC systems", 2022 DOE Exascale Computing Project Annual Meeting, April 2022
- [8] "Cache VOL: Efficient parallel I/O through caching data fast storage layer." HDF5 User Group meeting, October 2021
- [9] "Distributed Deep Learning", ALCF Simulation, Data and Learning Workshop, Argonne National Laboratory, October 2021
- [10] "Data Parallel Deep Learning", Argonne Training Program on Extreme-scale Computing (ATPESC) 2021, Aug 2021
- [11] "HDF5: Virtual Object Layer." 2021 Exascale Computing Project Community Birds-of-a-Feather Day, March 30, 2021 - April 1, 2021.
- [12] "Cache VOL: Efficient parallel I/O through caching data on node-local storage." 2021 DOE Exascale Computing Project Annual Meeting, April 12, 2021 - April 16, 2021.
- [13] "Distributed Deep Learning" and "profiling Deep Learning", ALCF Simulation, Data, and Learning Workshop, October 2020
- [14] "Dielectric dependent hybrid functionals for surfaces and interfaces", APS March Meeting, March 2019
- [15] "Dielectric properties of interfaces", APS March Meeting, Los Angeles, CA, March 2018
- [16] "Performance optimization of WEST and Qbox on Intel Knights Landing", IXPUG 2017 US Annual Meeting, Austin, TX, September 2017
- [17] "An introduction to quantum algorithms", Quantum Computing Tutorials, Argonne National Laboratory, December 2018
- [18] "Deep learning on Theta using TensorFlow, PyTorch, Keras, and Horovod", ALCF Simulation, Data, and Learning Workshop, Argonne National Laboratory, October 2018
- [19] "Porting Qbox to A21", ALCF Exascale application co-design meeting, Argonne National Laboratory, September 2018
- [20] "Quantum computing programing using ProjectQ", Quantum Computing Workshop, Argonne National Laboratory, July 2018
- [21] "Large scale electronic structure simulations on Theta", ALCF Simulation, Data, and Learning Workshop, Argonne National Laboratory, February 2018

Workshops and symposia organized

- [1] Microscopy & Microanalysis Symposium 2020, Section A01 – Advances in Modeling, Simulation, and Machine Learning for Microscopy Data sets, August 2020
- [2] ALCF Simulation, Data and Learning Workshop, October 2021
- [3] ALCF Simulation, Data and Learning Workshop, October 2019

Patents

- [1] Zheng, H., Schwartz, J., Jiang, Y., Hovden, R., et al. Recovering Atomic-Scale Chemistry From Fused Multi-Modal Electron Microscopy. U.S. Patent, granted by the U.S. Patent and Trademark Office, 2026.

Significant software developed or equipment designed

- [1] BlendCorpus: <https://github.com/zhenghh04/blendcorpus>
- [2] MLFlow: https://github.com/zhenghh04/alcf_mlflow
- [3] ClearML: https://github.com/zhenghh04/alcf_clearml_deployment
- [4] Agentic workflows: https://github.com/zhenghh04/amsc_agentic_workflows
- [5] HDF5 Cache VOL connector (main developer): <https://github.com/hpc-io/vol-cache>
- [6] DLIO Benchmark (main developer): https://github.com/argonne-lcf/dlio_benchmark

- [7] MLPerf Storage (main developer): <https://github.com/mlcommons/storage>
- [8] Boundary Element Method (BEM) (main developer), close source
- [9] Tomo_TV: https://github.com/jtschwar/tomo_TV
- [10] H5bench: <https://github.com/hpc-io/h5bench>
- [11] WEST: <http://www.west-code.org/people.php>
- [12] Qwalk: <https://github.com/QWalk/mainline>
- [13] Qbox: <https://github.com/qboxcode/qbox-public>

Major program initiatives and proposals funded

- [1] “American Science Cloud” – co-leading the Model Services area
- [2] “ExalO: ExalO - Delivering Efficient Parallel I/O on Exascale Computing Systems with HDF5 and UnifyFS”
 Role: Argonne Co-PI
 Agency: U.S. Department of Energy ASCR Exascale Computing Program
 Status: Awarded
 Period: 2019-2023
 Argonne Amount: \$1,232,329

Major projects managed or directed

- [1] Argonne Data Science Program (ADSP), 2020-2021, *Dynamic Compressed Sensing for Real-Time Tomographic Reconstruction*, Co-PI, 27k node-hours on Polaris @ ALCF
- [2] Argonne Data Science Program (ADSP), 2019-2020, *Dynamic Compressed Sensing for Real-Time Tomographic Reconstruction*, Co-PI, 16k node-hours on Summit @ OLCF
- [3] U.S. DOE ASCR Leadership Computing Challenge (ALCC) Award, 2017-2018, Computational Engineering of Electron - Vibration Coupling Mechanisms, Co-PI, 60 million hrs)
- [4] ALCF Director’s Discretionary (DD) Award, 2017-2018, Quantum Monte Carlo study of correlated electron materials, PI, 2.5 million hrs
- [5] Laboratory Computing Resource Center (LCRC) project, 2019
 Electronic structure of heterogeneous systems using dielectric dependent hybrid functionals, PI, 700 k hrs.

Mentorship

Postdoc mentored

- [1] Xiaolong Ma, 1/2026 – Present (Career Mentor)
- [2] Kaushik Velusamy, 1/2021 – 7/2024
- [3] Michele Morales, 3/2022 – 3/2024 (Career mentor)

Predoc mentored

- [1] Jingyang Jiang

Intern students mentored

- [1] Akhilesh Bondapalli, 2025 & 2026
- [2] Krishna Rauniyar, 2026
- [3] Jennifer Ha, John Wu, Colin Luangrath, Izzet Yildiri, 2024
- [4] Zhenghao Zhao, 2023
- [5] Zhao Zheng, 2022
- [6] Hariharan Deverajan, 2020
- [7] Sergio M. Servantez, 2019

Sustained interaction with sponsors or industry

- [1] Aurora Non Recurring Engineering – Co-lead for Deep Learning Frameworks

[2] MLCommons for MLPerf Storage Working Group work.